Protect information without reading it

Encode a state $\longrightarrow$ measure checks $\longrightarrow$ correct

The question for today

Can we learn how the information was disturbed without learning the information?

Final lecture: a visual tour of §§1–4 of the seminar notes.

One minute of quantum vocabulary

Vectors for calculations

$$\mathcal{H}_n = (\mathbb{C}^2)^{\otimes n}, \qquad \dim \mathcal{H}_n = 2^n.$$

$$u = \alpha|0\rangle + \beta|1\rangle.$$

Rays for pure states

$$[u] \in \mathbb{P}(\mathcal{H}_n), \qquad [u] = [\lambda u] \ (\lambda \neq 0).$$

$$[|0\rangle + |1\rangle] \neq [|0\rangle - |1\rangle].$$

Global phase disappears; relative phase remains

A code is $\mathbb{P}(\mathcal{C})$ for a linear subspace $\mathcal{C} \subseteq \mathcal{H}_n$.

Two errors, one measurement rule

$$X|b\rangle = |b+1\rangle, \qquad Z|b\rangle = (-1)^b|b\rangle, \qquad Y = iXZ.$$

Bit flip Phase flip

For a signed self-adjoint involution g:

$$P_\pm = \frac{I \pm g}{2}, \qquad \Pr(\pm \mid [u]) = \frac{\|P_\pm u\|^2}{\|u\|^2}.$$

An eigenstate is unchanged by this measurement. The sign of the lift labels the sectors.

Repetition: measure the parity, preserve the superposition

$$\alpha|0\rangle + \beta|1\rangle \longmapsto u = \alpha|000\rangle + \beta|111\rangle.$$

$$g_1 = Z_1 Z_2, \qquad g_2 = Z_2 Z_3.$$

error	I	X_1	X_2	X_3
syndrome	00	10	11	01

The outcome depends on the error, not on α, β.

$Z_1 u = \alpha|000\rangle - \beta|111\rangle$ is invisible to both checks.

Stabilizers: specify the code by equations

Definition

Choose an abelian group S of Pauli matrices with $-I \notin S$. The underlying code subspace is

$$\mathcal{C}(S) = \{u : gu = u \text{ for every } g \in S\}.$$

$$\underbrace{g_1 u = u, \quad \ldots, \quad g_r u = u}_{\text{commuting signed checks}}$$

The code is the selected common $+1$ sector, not the entire projective fixed locus.

How much information fits?

n physical qubits $\qquad$ r independent checks

$$\dim \mathcal{C} = 2^{n-r} = 2^k.$$

$$k = n - r \text{ encoded qubits}$$

Repetition: $n = 3$, $r = 2$, $k = 1$.

Count independent checks. Redundant measured checks do not remove more dimensions.

A syndrome is a list of violated equations

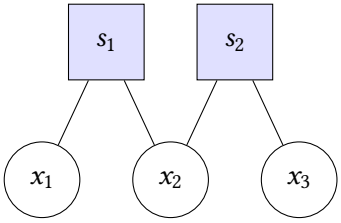
$$g_j E = (-1)^{s_j(E)} E g_j \qquad \Longrightarrow \qquad g_j(Eu) = (-1)^{s_j(E)} Eu.$$

$$0 \leftrightarrow +1 \qquad\qquad 1 \leftrightarrow -1$$

What the checks tell us

Which sector contains the corrupted state. They need not identify the actual error uniquely.

A picture of two parity equations



$$x_1 + x_2 = s_1$$

$$x_2 + x_3 = s_2$$

All sums are in $\mathbb{F}_2$.

Set $x = (0, 1, 0)$: both checks light up.

Read the edges first. Each check adds the variables attached to it.

The picture has a name: Tanner graph

Definition

For a binary matrix H, use a variable vertex for each column and a check vertex for each row. Join them exactly when $H_{aj} = 1$.

$$H = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad s = Hx.$$

In a CSS code, two matrices describe the checks:

$$H_X H_Z^{\mathsf{T}} = 0, \qquad s_Z = H_Z x, \quad s_X = H_X z.$$

X checks detect Z components; Z checks detect X components. Y has both.

Why the checks can coexist

Different types commute when they overlap evenly.

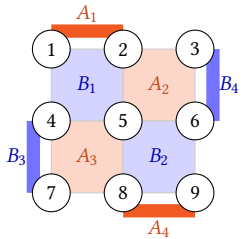
$$\underbrace{X_2X_3X_5X_6}_{A_2} \qquad \underbrace{Z_1Z_2Z_4Z_5}_{B_1}$$

Two anticommutations: qubits 2 and 5.

$$(-1)^2 = +1.$$

The equation $H_X H_Z^{\mathsf{T}} = 0$ says this for every pair of checks.

A complete quantum code on nine qubits



Orange: X checks A_j.

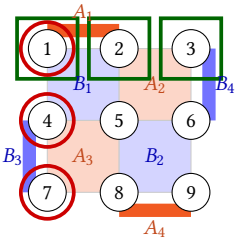
Blue: Z checks B_j.

A square acts on its four corners; a boundary bar on its endpoints.

$$n = 9, \qquad r = 8, \qquad k = 1.$$

No extra data qubits at the face centers. Qubits are numbered in row order.

Invisible does not mean harmless



$$\overline{X} = X_1 X_4 X_7, \qquad \overline{Z} = Z_1 Z_2 Z_3.$$

Each commutes with every check.

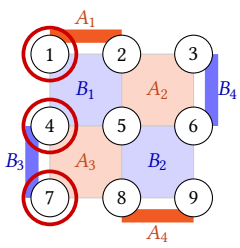
They overlap once and anticommute with each other.

Logical operators

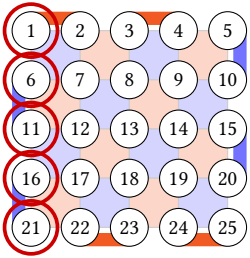
They preserve the code subspace but can change the encoded state.

Red rings: X support. Green boxes: Z support. A logical error may fix some code states.

Distance: make harmful invisible operators large



$[[9, 1, 3]]$



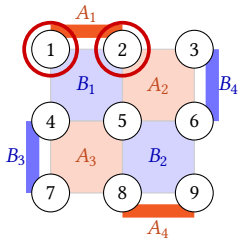
$[[25, 1, 5]]$

$[[n, k, d]]$

n physical qubits, k encoded qubits, d minimum weight of a nontrivial logical Pauli.

Odd rotated patches are $[[d^2, 1, d]]$. Distance d guarantees correction when $2t < d$.

Two different errors can be equivalent



$$A_1 = X_1 X_2.$$

For a code vector u, $A_1 u = u$, so

$$X_1 u = X_2 u.$$

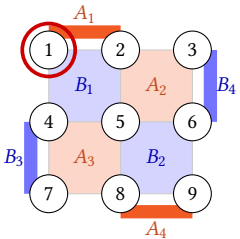
Degeneracy in this example

The errors have the same syndrome and the same action on the code. Either correction can succeed.

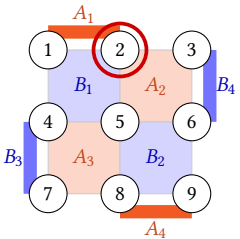
A stabilizer changes the representative error, while preserving its effect on encoded states.

Three corrections: ask the audience

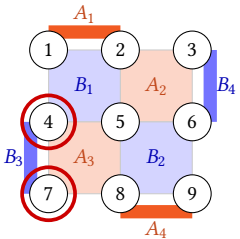
Actual error: X_1. Only B_1 changes sign.



Correction X_1
Residual I



Correction X_2
Residual $X_1X_2 = A_1$



Correction X_4X_7
Residual $\overline{X}$

All restore the checks. Which restore the encoded information?

The decoder must choose the right equivalence class

For an X error x, choose a correction $\widehat{x}$.

Consistency

$$H_Z\widehat{x} = H_Z x \quad \Longleftrightarrow \quad x + \widehat{x} \in \ker H_Z.$$

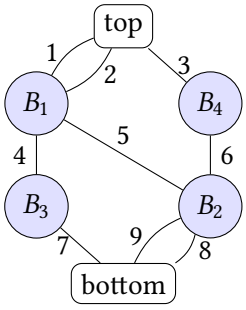
Success

$$x + \widehat{x} \in \mathrm{row}(H_X).$$

Residual logical classes: $\ker H_Z / \mathrm{row}(H_X)$.

This is ordinary binary linear algebra. The probability model decides which class to choose.

Redraw the same patch as a decoding graph



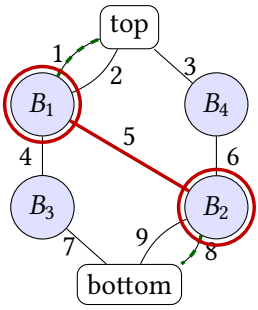
Vertices: Z checks and two boundaries.

Edges: data qubits, labeled $1, \ldots, 9$.

An X error is an edge set. Its syndrome is the set of odd-degree check vertices.

This is not the Tanner graph: variables have become edges. Boundaries carry no parity constraint.

Decoding becomes pairing endpoints



Defects at B_1, B_2:

$$\{5\} \qquad \text{or} \qquad \{1, 8\}.$$

Solid: edge 5. Dashed: edges 1 and 8.

Connect two defects, or connect each to a boundary.

Minimum-weight matching chooses a cheapest collection of paths.

For equal independent bit-flip probabilities below 1/2, cheapest means fewest edges.

A noise model changes the geometry

Independent X errors, with $\Pr(x_j = 1) = p_j < 1/2$:

$$\Pr(x) = \prod_j p_j^{x_j} (1 - p_j)^{1 - x_j}.$$

$$\text{Most likely error} \iff \min_{H_Z x = s} \sum_j w_j x_j, \qquad w_j = \log \frac{1 - p_j}{p_j}.$$

$p_1 = p_8 = 0.1$, $p_5 = 0.01$:

$$w_1 + w_8 = 2 \log 9 < \log 99 = w_5.$$

Two likely faults can be more plausible than one very unlikely fault.

Most likely error versus most likely class

A small CSS example: $H_X = (1\ 1\ 0)$, $H_Z = (1\ 1\ 1)$. Priors $(0.2, 0.2, 0.3)$; observed syndrome 1.

class	individual unnormalized masses	total
$\{100, 010\}$	$0.112 + 0.112$	$\mathbf{0.224}$
$\{001, 111\}$	$\mathbf{0.192} + 0.012$	0.204

Matching optimizes an error.
Optimal quantum recovery chooses a class by its total probability.

This toy is $[[3, 1, 1]]$. It illustrates inference, not useful protection against arbitrary faults.

What scaling up buys us

More qubits $\longrightarrow$ longer logical strings

A threshold question

For a growing code family, a fixed noise model, and a specified decoder, does logical failure probability tend to zero as the code grows?

For surface codes with independent errors and suitable decoding: yes, when the error probability is sufficiently small.

Today: perfect checks and X errors. Faulty repeated measurements require an additional time direction.

One picture to take away

An unknown quantum state

$\downarrow$ commuting parity measurements

A classical syndrome

$\downarrow$ inference modulo stabilizers

A correction that preserves the encoded information

A consistent correction can still be a logical error. That is the decoder's central difficulty.

Continue exploring

Seminar notes: §§1–4; later sections develop BP, qLDPC and circuit noise.

Interactive patch: toggle errors and compare corrections (linked on the talk page).

References

D. Gottesman, *An introduction to quantum error correction and fault-tolerant quantum computation*, arXiv:0904.2557, §§2–3.
E. Dennis et al., *Topological quantum memory*, J. Math. Phys. 43 (2002).
J. Roffe, *Quantum error correction: an introductory guide*, Contemp. Phys. 60 (2019).

`https://utsa-quantum-math.github.io/talks/2026-11-13-duenez/`